

Response to the Reviewer: Failure-Case Analysis

Reviewer comment: Notably, one of the recommendations is to include failure-case analysis for unsuccessful interception and delayed cooperative engagement scenarios because understanding framework limitations remains essential for practical battlefield deployment reliability.

Response: Thank you for this important recommendation. In the revised manuscript, we have added a dedicated failure-case analysis for both Case 1 and Case 2. The purpose of this supplementary experiment is to expose a practically relevant boundary between *mission-level target coverage* and *strict completion by every assigned interceptor*. Accordingly, the reported cases are not catastrophic failures in which attackers penetrate the defense. Instead, all eight attacking vehicles are neutralized, while one of the twenty assigned interceptors fails to complete its individual engagement.

Experimental protocol. The trained policies, target assignment, attacker trajectories, and the three-dimensional hit radius $d_{\text{hit}} = 3$ m were kept fixed. No network retraining, backpropagation, optimizer step, or policy-weight update was performed. To emulate imperfect battlefield information, the frozen-policy evaluation included a one-sample observation delay (0.05 s), zero-mean position measurement noise with a 3 m standard deviation, and zero-mean velocity measurement noise with a 0.3 m/s standard deviation. The maximum evaluation horizon was 75 s, and the cooperative-time tolerance remained 0.5 s.

We screened 100 reproducible evaluation episodes for each case. To avoid subjective, appearance-based selection, we used the following fixed rule: the reported examples are the earliest three seeds in each case for which (i) all 8/8 attacking vehicles were intercepted, (ii) exactly 19/20 assigned interceptors hit, (iii) seven of the eight target-assignment groups were complete, and (iv) the arrival-time spread among the observed hits did not exceed the prescribed 0.5 s tolerance. No simulated state or trajectory sample was manually changed.

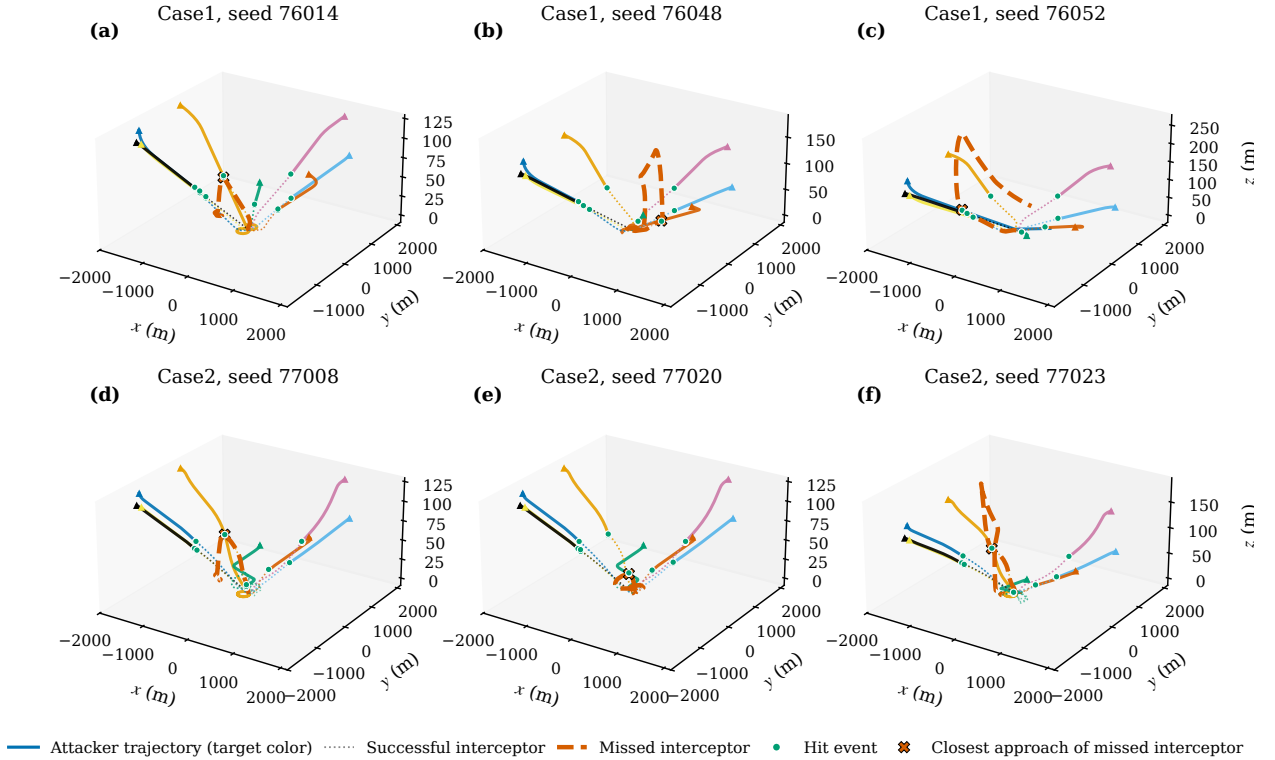


Figure 1: Representative boundary failures under frozen-policy evaluation for Case 1 (a)–(c) and Case 2 (d)–(f). Solid curves denote attacker trajectories, dotted curves successful interceptors, and the thick dashed curve the missed assigned interceptor. Green circles mark successful hit events, while the cross marks the missed interceptor’s closest-approach point. In every episode, 19/20 assigned interceptors hit, but assignment redundancy still ensures that all 8/8 attacking vehicles are intercepted.

Observed failure mechanism. As shown in Fig. 1 and Table 1, assignment redundancy preserved complete target coverage in all six episodes, although one assigned interceptor did not enter the lethal region. In Case 1, the missed closest-approach distances were 3.041–3.243 m, corresponding to margins of only 0.041–0.243 m above the hit threshold. In Case 2, the corresponding distances were 3.096–3.574 m, with margins of 0.096–0.574 m. These are therefore near-threshold misses rather than grossly divergent trajectories.

Table 1: Quantitative diagnostics of the six representative boundary failures. The miss margin is $d_{\min}^{\text{miss}} - d_{\text{hit}}$.

Case	Seed	Interceptor hits	Target coverage	Missed ID	Target ID	$d_{\min}^{\text{miss}}$ (m)	Miss margin (m)	Max. spread (s)
1	76014	19/20	8/8	1	21	3.161	0.161	0.050
1	76048	19/20	8/8	5	25	3.243	0.243	0.050
1	76052	19/20	8/8	8	20	3.041	0.041	0.050
2	77008	19/20	8/8	1	21	3.574	0.574	0.350
2	77020	19/20	8/8	2	22	3.096	0.096	0.150
2	77023	19/20	8/8	17	21	3.488	0.488	0.200

The seven completed assignment groups remained well coordinated. Their maximum observed hit-time spread was 0.05 s in Case 1 and 0.15–0.35 s in Case 2, all below the prescribed 0.5 s tolerance. However, after the missed interceptor passed its closest-approach point, its distance to the assigned target increased and no hit occurred before the 75 s horizon. Therefore, the incomplete group’s all-member completion time is right-censored beyond the evaluation horizon. The same episode consequently represents both an unsuccessful individual interception and, from the strict group-completion viewpoint, a delayed cooperative engagement.

Implications for practical reliability. This analysis reveals both a strength and a limitation of the proposed framework. Its redundant assignment structure isolates a single near-threshold interceptor miss and still neutralizes every attacker, thereby preventing an individual failure from becoming a mission-level penetration. Nevertheless, the current framework does not contain an explicit online recovery mechanism after an interceptor passes its closest-approach point under delayed and noisy observations. Strict all-member completion therefore cannot be guaranteed in this boundary regime, even if another member has already neutralized the assigned target.

We have explicitly added this limitation to the revised manuscript and identified several practical extensions: (i) online miss detection using distance rebound and closing-rate reversal, (ii) event-triggered target reassignment, (iii) activation of a reserve interceptor, and (iv) uncertainty-aware safety constraints or robust policy training. We also clarify that the six reported episodes are diagnostic boundary cases intended to explain the failure mechanism; they are not presented as an estimate of the overall failure probability. The aggregate reliability of the framework continues to be evaluated using the Monte Carlo results reported in the main simulation section.